



Artificial intelligence and numerical weather prediction models: A technical survey

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ABSTRACT

Can artificial intelligence (AI) models beat traditional numerical weather prediction (NWP) models based on physical principles? The rapid advancement of AI, inherent computational limitations of NWP models, and the lack of access to big data drive this question in terms of resolution and complexity. This survey offers a systematic review of studies that integrate AI with NWP models at various stages of weather and climate modeling. It aims to address key research questions, including the types of forecasting models, the integration of AI into NWP systems, and the comparative efficacy of AI-based approaches versus conventional NWP models. It covered peer-reviewed literature from 2000 to 2024. This technical survey highlights key advancements in the application of AI within NWP modeling in data assimilation, augmentation, pre-processing, adaptive parameter tuning, optimization, uncertainty quantification, extreme event prediction, post-processing, and the interpretation of NWP outputs. While AI demonstrates significant potential in post-processing NWP outputs, pre-processing remains challenging. This survey also presents state-of-the-art AI-based hybrid models and assesses their applicability to weather data. It highlights the promise of AI in potentially replacing traditional NWP models but emphasizes the need for further advancements in model development and application. The study also offers a detailed classification of forecasting models and outlines promising directions for future research.

1. Introduction

Weather and climate prediction have been fundamental throughout history (Chen et al., 2023). Weather forecasts are essential in decision-making and safety measures to mitigate exposure to hazardous conditions and manage health risks during extreme temperatures (Abbe, 1901). Weather-related variables, such as temperature, humidity, precipitation, and wind dynamics, significantly impact various aspects of human livelihoods (Ren et al., 2021). These variables also provide critical insights to tackle severe weather events, i.e., floods or droughts (Zheng et al., 2014). In the early 20th century, weather forecasting was highly imprecise, relying on sparse and irregular observations with limited theoretical physics applications. Forecasters used extrapolation, local climatology, and intuition, plotting weather variables on maps to identify patterns like depressions and anticyclones (Bhawsar et al., 2021). Forecasting heavily depended on empirical rules and convection processes, but the non-linear nature of convection posed significant challenges beyond human intuition (Lynch, 2008; Waqas et al., 2024a). Weather

forecasting techniques have evolved from statistical methods before 2000 to advanced process-based statistics and the Internet of Things (IoT) (Waqas et al., 2023). Waqas et al. (2023) stated the development across three periods: pre-2000, 1941–2000, and the current era (Waqas et al., 2023). Between 1900 and 2000, advancements included stochastic methods, power regression models, and Numerical Weather Prediction (NWP) models facilitated by real-time data and supercomputing (Ashok et al., 2022). The first attempt at manual NWP was conducted by Lewis Fry Richardson in Britain in 1922, with the advent of computer-assisted weather forecasts in 1950 (Lynch, 2008). This development was soon followed by the implementation of operational weather forecasting systems in Sweden, the USA, and Japan (Schultz et al., 2021). NWP represents the forefront of weather forecasting. It applies partial differential equations to model atmospheric dynamics based on initial conditions, integrating small-scale processes through physical parameterizations. Unresolved phenomena like cloud microphysics and turbulence are approximated within the model framework (Pu et al., 2019). After 2000, advancements in weather observations have enhanced NWP model

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accuracy. The deployment of low-Earth-orbit and geosynchronous satellites, which now provide 85% of data for global weather models, alongside private investments, has expanded observational capabilities (Bluestein et al., 2022).

Additionally, the IoT and autonomous platforms improved local observations (Brotzge et al., 2023). Data assimilation (DA) techniques have evolved, with methods like the ensemble–variational (EnVar) approach enhancing the integration of observations and forecasts (Lorenc, 2014; Aulign et al., 2016). The Joint Effort for Data Assimilation Integration (JEDI) enhances community engagement in DA research and development (Tré et al., 2020). Advances in model dynamics, parameterizations, and ensemble prediction extended forecast ranges and improved accuracy, with tools like the extreme forecast index (EFI) aiding in extreme event preparation (Liguori et al., 2023). Despite the advancements in NWP modeling, challenges persist—for example, model accuracy limitations, high numbers of supported models, and difficulties assimilating new data.

The US global NWP, while competitive, faces issues with rare event forecasting and model error representation. Data volume, gaps, resource constraints, and collaboration limitations hinder progress. These issues required improved high-performance computing capacity, enhanced data assimilation methods, and a strategic approach to resource allocation (Brotzge et al., 2023). These challenges pushed scientists and researchers to develop smart solutions to address high data requirements and computational demands of NWP models. During the 1990s–2000s, advancements in Artificial Intelligence (AI) through the development of Machine Learning (ML) and Deep Learning (DL) models mitigated these challenges and enhanced computational efficiency and accuracy in weather forecasting. After 2010, three key advancements—massively parallel processing in graphical processing units (GPUs), efficient analysis of large datasets via convolutional neural networks (CNNs), and the availability of benchmark datasets online—ignited the wave of AI, leading to breakthroughs in image recognition (Hijazi et al., 2015), speech recognition (Abdel-Hamid et al., 2014), weather-related analysis through complex neural networks (Hewage et al., 2020). With these developments, many researchers have developed hybrid models by integrating ML/DL models with NWP for different stages and propose for weather forecasting. This integration has enhanced the precision and efficiency of weather predictions. These models enable hybrid modeling approaches by (i) combining ML/DL with WRF for hybrid modeling, (ii) improving data pre-processing for WRF, and (iii) post-processing techniques to refine outputs. These models are also significant in (iv) predicting extreme weather events, (v) adjusting and correcting WRF outputs, and (vi) downscaling climate projections for enhanced spatial resolution. Moreover, they facilitate (vii) quantification of uncertainty and (viii) dynamic tuning of WRF parameters, which advanced the accuracy and reliability of weather forecasting models.

Numerous studies have explored the potential of advanced techniques to enhance traditional forecasting models. For instance, Ayoub et al. (2024) developed statistical emulators that integrate deep neural networks (DNNs) with vector autoregressive models to predict wind and temperature fields, facilitating real-time forecasts crucial for responding to nuclear accidents (Ayoub et al., 2024). Zhang et al. (2024) proposed a Population-Based Training Gated Recurrent Unit (PBT-GRU) model to adjust WRF precipitation forecasts, which led to a 63% decrease in root mean square error (RMSE) compared to traditional forecasting methods (Zhang et al., 2024). Furthermore, Mu et al. (2023) introduced the Fortran Torch Adaptor to integrate DL emulators with the WRF model to increase forecast accuracy and computational efficiency during extreme weather scenarios (Mu et al., 2023). Another study utilized a CNN to bias-correct WRF outputs, resulting in better alignment across various meteorological parameters (Sayeed et al., 2021). Ghazvinian et al. (2024) employed a DL model to improve probabilistic quantitative precipitation forecasts from a 200-member ensemble WRF model, demonstrating enhanced skill in predicting extreme precipitation events (Ghazvinian

et al., 2024). Many review studies (Chen et al., 2023; Ren et al., 2021; Bhawsar et al., 2021; Lynch, 2008; Waqas et al., 2023; Schultz et al., 2021; Jaseena et al., 2022; Wu et al., 2024) have addressed the ML and DL techniques or integration with NWP models for weather forecasting.

Reviewed studies highlight that the integration of AI offers an advancement in increasing the accuracy, efficiency, and adaptability of NWP models. Unlike traditional methods that rely on physical principles, AI models learn complex patterns from vast datasets, making them effective in post-processing, uncertainty quantification, and extreme event prediction. This distinction underscores the need to explore AI's potential in addressing the limitations of traditional NWP systems. However, a critical gap exists in evaluating the implementation phases and objectives associated with ML/DL models when combined with NWP models, as well as the advancements achieved through such integrations. This review addresses the main research question: Are ML and DL models superior to NWP models? The primary objective is to technically evaluate and understand the integration of ML and DL techniques with NWP models, assessing improvements in hybrid modeling, data pre-processing, post-processing techniques, and predictions of extreme weather events. The focus is on how this integration enhances weather forecasting accuracy and operational efficiency.

2. Materials and methods

A systematic review of recent advancements in the integration of NWP models with ML and DL techniques was conducted for this study. Initially, research questions (RQs) were formulated based on selected keywords, guiding the selection of publications from 2000 to 2024, as illustrated in Fig. 1. Screening criteria were established to identify relevant studies and systematically address the research questions. A detailed explanation of the methodology is provided in the following subsections.

2.1. Selection and screening criteria

2.1.1. Research questions

Based on the limitations identified in the existing literature and the objectives of this study, the following research questions (RQs) have been formulated:

RQ-1: What types of weather-related forecasting models are used in the literature?

RQ-2: What are the advancements, challenges, and opportunities of NWP models with ML/DL integration?

RQ-3: How and at what phase of modeling machine and deep learning algorithms integrate with numerical weather prediction models?

RQ-4: Can end-to-end AI-based forecast weather be better than NWP models?

RQ-5: What are the future recommendations for integrating AI and NWP models?

2.1.2. Keywords selection

This process involves formulating the query string by searching the literature for relevant keywords, which were then organized according to a predefined sequence to construct the final query. The generated query includes specific keywords such as “Weather Forecasting Models,” “Numerical Weather Prediction (NWP) Models,” “Deep Learning and Machine Learning in Weather Prediction,” “Deep Learning Integration with NWP,” “Standalone Machine and Deep Learning Models.”

2.1.3. Domain selection

Different databases and search engines, including ScienceDirect, MDPI, Google Scholar, ACM Digital Library, SpringerLink, IEEE Xplore, and Web of Science, are used. The selection criteria focused on articles published in reputable, peer-reviewed journals to ensure the inclusion of high-quality and impactful research.

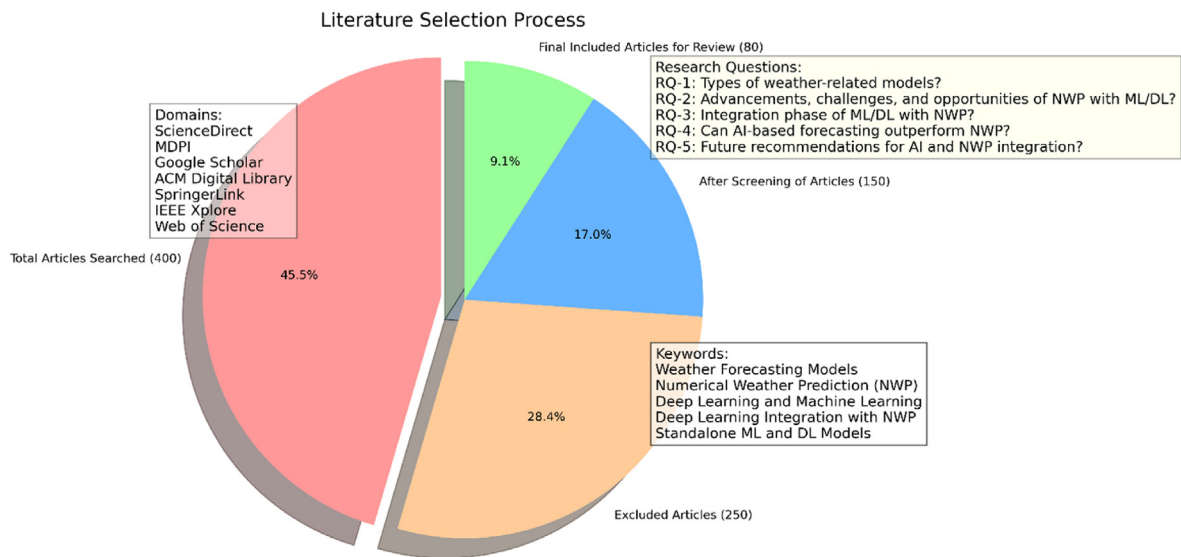


Fig. 1. Literature selection process.

2.1.4. Inclusion Criteria

This study follows the following Inclusion Criteria: (1) Publications from 2000 to 2024; (2) Publicly available journal articles, conference, or workshop proceedings; (3) Publications that used ML and DL with NWP and alone for weather-related forecasting; (4) Studies comparing data mining and forecasting strategies; (5) Publications presenting improved or customized methodologies for weather prediction; (6) Articles integrating AITs with other techniques.

2.1.5. Exclusion criteria

Exclusion norms include: (1) Non-English documents; (2) Articles published outside 1990–2024, except foundational studies; (3)

Publications not focused on rainfall forecasting; (4) Studies not utilizing weather observation data; (5) Articles not evaluating the performance of proposed methods.

2.1.6. Literature evaluation criteria

The following evaluation procedures were implemented to ensure high-quality outcomes: (1) Research articles were sourced from credible and authoritative online libraries; (2) Recent publications were prioritized to ensure findings reflect the latest advancements; (3) The selection process was conducted impartially; and (4) Each step of the review was executed with meticulous attention to detail and accuracy.

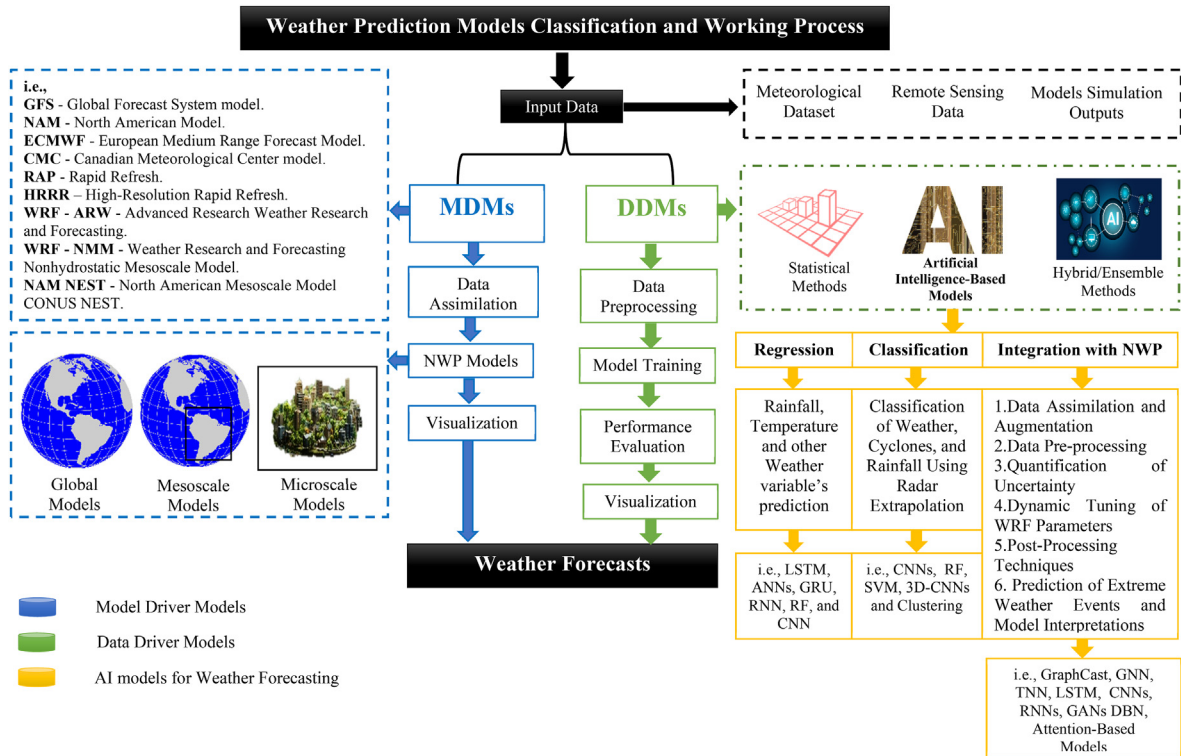


Fig. 2. Weather prediction models classification and process of weather forecasting.

3. Classification of weather forecasting models

Traditionally, weather forecast models are classified into model-driven models (MDMs) and data-driven models (DDMs), as shown in Fig. 2 (Waqas et al., 2024b). MDMs are physics-based subjects emphasizing improving forecast accuracy by comprehending fundamental atmospheric processes. However, the rapid expansion of multisource and multi-dimensional meteorological data has transformed weather prediction into a complex spatio-temporal data challenge. DDMs such as statistical, AI, and hybrid models have recently emerged to address these challenges. DDMs addressed the challenges of traditional MDMs by revealing complicated spatial-temporal correlations and advanced DL and ML algorithms (Waqas et al., 2024b).

3.1. Numerical weather prediction model

The NWP models use complicated mathematical equations reflecting atmospheric dynamics and thermodynamics to forecast weather conditions based on preliminary atmospheric data (Bauer et al., 2015; Richardson, 1922). The general working mechanism of the basic NWP model is shown in Fig. 3. Continuous improvements in their predictive skills have marked the progress of NWP models (Lynch, 2008). This progress is driven by ensemble forecasting, advanced data assimilation, and improved representation of physical processes like convection, radiation, and boundary layer dynamics. Ensemble forecasting allows for the integration of multiple simulations to assess forecast uncertainties, providing probabilistic forecasts rather than deterministic ones (Pu et al., 2019). This method enhances the reliability of forecasts, particularly for extreme weather events such as hurricanes and heat waves (Bauer et al., 2015).

The development of high-resolution global models running at scales close to 1 km is one of the latest advancements in NWP. These models promise to reduce the need for parameterizations of small-scale processes, allowing for a more explicit representation of atmospheric dynamics (Dudhia, 2014). However, this resolution requires significant computational power and poses a considerable challenge due to the complex processes at these fine scales (Banadkooki et al., 2019).

Opportunities for further advancements in NWP are abundant (Brotzge et al., 2023). The integration of more sophisticated data assimilation techniques, such as four-dimensional variational data assimilation (4D-Var) and ensemble-based methods, can significantly improve the initialization of models, leading to better forecast accuracy (Zhang et al., 2012; Maulik et al., 2022). Additionally, coupling atmospheric models with the oceanic, land surface, and cryosphere models opens new opportunities for comprehensive Earth system modeling (Gettelman et al., 2022). Such coupled models can enhance predictive skills across various time scales, from short-term weather forecasting to seasonal climate predictions (Wang et al., 2022a).

Despite the progress and opportunities, NWP models face several challenges. The representation of physical processes at sub-grid scales remains a fundamental difficulty. Parameterizations of convection, cloud microphysics, and surface interactions introduce uncertainties that can propagate and amplify within the forecast (Brotzge et al., 2023). Another major challenge is the computational demand for high-resolution models. Running global simulations at kilometer-scale resolution requires immense computational resources, posing limitations on operational forecasts' spatial and temporal resolution. Furthermore, the integration of complex data, including satellite and ground-based observations, into models demands continuous advancements in data assimilation techniques to handle large volumes and diverse types of data efficiently (Bauer et al., 2015). Addressing these challenges will be critical to the future evolution of NWP models.

3.2. Statistical methods

Statistical methods, such as Autoregressive Moving Average (ARMA), Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving Average (SARIMA), multiple regression, and Vector Autoregression (VAR), are linear models used to forecast short-term weather based on past data (Athanasiadis et al., 2006; Abraham et al., 2009). Linear regression, which includes primary and multiple variations, models linear relationships between variables and uses gradient descent to reduce prediction errors (Montgomery et al., 2021). ARIMA uses time series data to forecast future observations, whereas

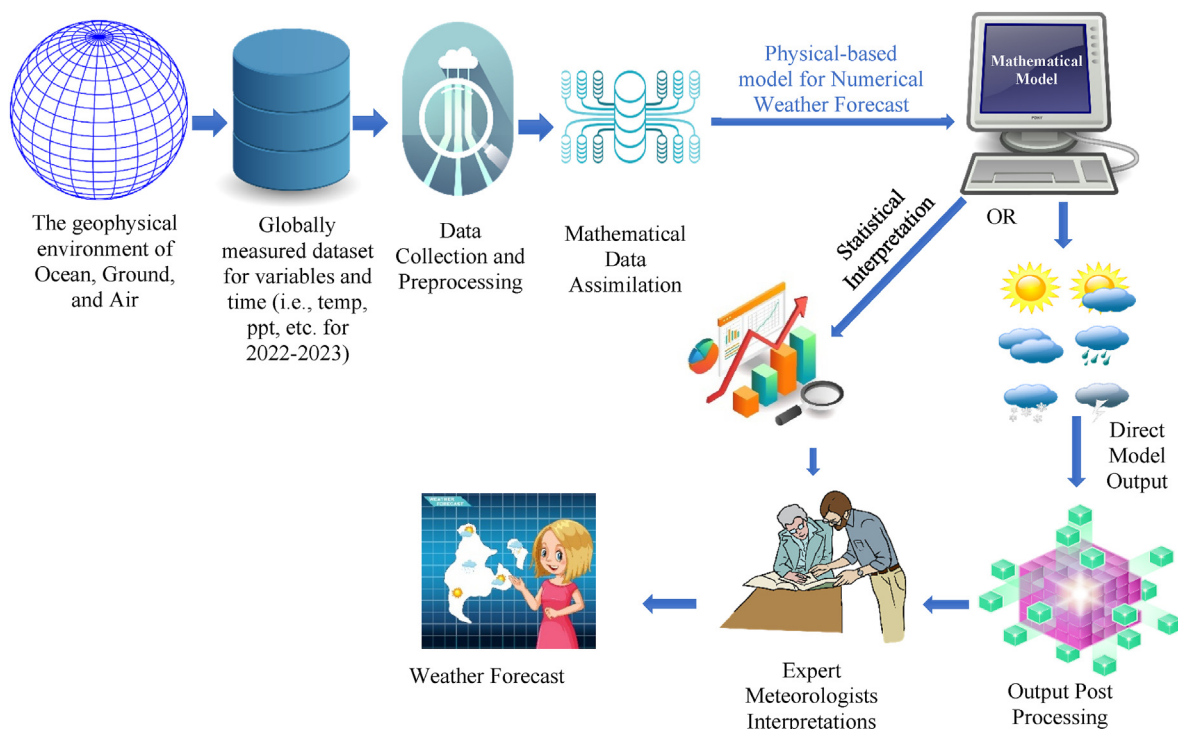


Fig. 3. General working process of numerical weather prediction model.

VAR, a multivariate extension of autoregression, works well for related variable sets (Thomakos et al., 2004). Khashei and Bijari (2011) suggested hybrid models that combine ARIMA with ANN or other non-linear methods to improve prediction performance (Khashei et al., 2011). Similarly, many researchers have investigated statistical models for forecasting wind speed (Yousuf et al., 2022), power, and rainfall (Vil-larini et al., 2012), confirming their effectiveness across various meteorological applications. Despite the advantages of these models, they struggle with capturing complex, non-linear relationships and are limited in handling large-scale, multi-dimensional data, reducing their predictive accuracy in dynamic weather systems.

3.3. Machine and deep learning models

The Advanced ML and DL algorithms have emerged in the 21st century as robust models for weather-related forecasting. This study critically explores the applications of ML and DL techniques in agriculture, highlighting key works that showcase the essential features of these models. A detailed classification of AI, ML, and DL models, as explained by Waqas et al. (2023), is depicted in Fig. 4 (Waqas et al., 2023). ML models are categorized into four types based on the type of input or feedback received: supervised learning, where inputs are paired with desired outputs to learn mapping rules (Nasteski, 2017); unsupervised learning, which identifies hidden structures in data without labels (Adadi, 2021); semi-supervised learning, which combines labeled and unlabeled data, as seen in Generative Adversarial Networks (Ouali et al., 2020); and reinforcement learning, where models learn through interactions with dynamic environments (Li et al., 2022). The advancement of ML and DL's success is led by high computational power, massive datasets, and the rapid growth of ANN architectures after the 1990s (Sengupta et al., 2020). These advanced neural networks, such as CNNs, excel in extracting hierarchical characteristics from gridded data, making them useful in weather and climate applications, such as satellite image analysis (Deeprathi et al., 2021). Long short-term memory (LSTM) and gated recurrent units are advanced variations of recurrent neural networks (RNNs) built for sequential input. These are combined into complicated architectures like ConvLSTM for spatiotemporal applications (Sahoo et al., 2019; Yang et al., 2019; Hochreiter, 1998; Cho et al., 2022a; Le et al., 2019; Pinthong et al., 2023; Abotaleb et al., 2024). Emerging DL models like variational auto-encoders (VAE) (Chira et al., 2022) and generative adversarial networks (GAN) are generative models that learn data distributions to create new samples relevant to tasks, including image recognition (Alqahtani et al., 2021). These advanced architectures, which include numerous layers and massive DL networks, address complicated issues similar to NWP models (Laloyaux et al., 2022). Meteorologists have traditionally used ML models such as

regression and data assimilation, but newer DL approaches, including ANNs, provide better forecasting capabilities (Huntingford et al., 2019).

4. Integration of ML and DL with NWP at different stages

This section discussed the integration of ML and DL algorithms with the NWP model. This hybrid approach mitigates biases and uncertainties inherent in WRF outputs, improving prediction accuracy. Various algorithms, i.e., RF, SVM, BiLSTM, ConvLSTM, LSTM, CNNs, and GRUs, have demonstrated efficacy in historical data to refine real-time WRF forecasts. The combination of WRF with ML/DL techniques exploits the strengths of MDMs and enhances DDM's predictive capabilities. Critical aspects of this integration include feature engineering, pre-processing, post-processing, and data assimilation and augmentation, all of which are essential for optimizing the performance of the hybrid model. These methodologies contribute to the enhanced forecasting of extreme weather events, improved uncertainty quantification, and adaptive parameter tuning for reliable and accurate meteorological predictions. This section studies literature related to the application of ML and DL algorithms with NWP, as shown in Table 1.

4.1. Data assimilation and augmentation

Data assimilation is an integration of observational data with model forecasts (Rabier, 2005) (i.e., assimilation of satellite dataset in NWP model (Eyre et al., 2022)). Traditional assimilation methods used statistical methods such as Kalman filters, which, while effective, are computationally intensive and have issues understanding complex non-linear dynamics (Carrassi et al., 2018). Current advancements in ML and DL algorithms provide more efficient and robust alternatives. DNNs map observational data directly to model updates, which is more effective than statistical methods (Arcucci et al., 2021). Adaptive Gaussian process-based iterative ensemble smoother model uncertainty in observations and forecasts for more reliable data assimilation (Ju et al., 2018).

Furthermore, multi-dimensional CNNs are utilized to process image-based observational data to extract pertinent features for assimilation in tropical cyclone intensity estimation (Lee et al., 2019a). RNNs are adept at handling time-series data for atmospheric noise removal from InSAR time series with missing values, including radar observations. It captured temporal dependencies and handled missing values effectively (Zhao et al., 2021). TCNs also showed effective data assimilation for weather forecasting using time-series data from the local weather station (Hewage et al., 2020).

Data augmentation involves creating additional training data from existing datasets (Mumuni et al., 2022). Augmenting the diversity of training datasets plays a pivotal role in enhancing model generalization

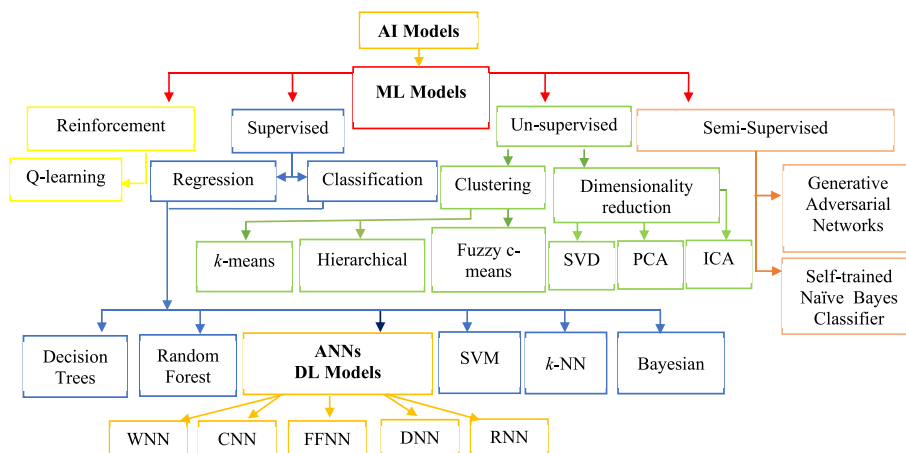


Fig. 4. Classification of artificial intelligence including ML and DL models (Waqas et al., 2023).

Table 1
NWP model stages and integration of ML and DL models reviewed studies.

Integration stages	Studies Reviewed
1. Data Assimilation and Augmentation	(Schultz et al., 2021; Hewage et al., 2020; Rabier, 2005; Eyre et al., 2022; Carrassi et al., 2018; Arcucci et al., 2021; Ju et al., 2018; Lee et al., 2019a; Zhao et al., 2021; Mumuni et al., 2022; Ge et al., 2024; Wangwongchai et al., 2023; Feng et al., 2021; Sun et al., 2023; Yan et al., 2023; Liguori et al., 2021; Rasheed et al., 2019, 2020)
2. Data Pre-processing	(Gough et al.; Fay et al., 1997; Mathiesen et al., 2013; Wang et al., 2021; Senocak et al., 2023; Georgakaki et al., 2024; Micolini et al., 2020; Abdulla et al., 2022; Maraun et al., 2010; Li et al., 2021; Al- et al., 2024; Yan et al., 2021; Shen et al., 2024; Garcia et al., 2018; Akbarian et al., 2023; Phan et al., 2021; Gu et al., 2019; Jia et al., 2021),
3. Quantification of Uncertainty	(Qian et al., 2016; Mehra, 2021; Speich et al., 2021; Mitros et al., 2019; Williams et al., 2006; Breiman, 2001; Gal et al., 2016; Tang et al., 2022; Cheng et al., 2023)
4. Dynamic Tuning of WRF Parameters	(Ollinaho et al., 2013; Loh et al., 1999; Ting et al., 2009; Gao, 2003; Koster et al., 2018; Sward et al., 2022; Almaraashi, 2023; Zhang et al., 2015, 2020, 2022a; Couvreur et al., 2021; Christianson et al., 2023; Menon et al., 2020; Mohammadi et al., 2020; Mnih et al., 2015; Finn et al., 2017)
5. Post-Processing Techniques	(Mehra, 2021; Buizza, 2018; Grö et al., 2021; Baran et al., 2018; Duan et al., 2021; Hamill et al., 2007; Rasp et al., 2018; Brenowitz et al., 2018; Baran et al., 2021; Monego et al., 2022; Dong et al., 2023; Lee et al., 2019b; Evin et al., 2021; Ahmed et al., 2020; Scheuerer et al., 2020; Ghazvinian et al., 2022; Wang et al., 2024; Price et al., 2022; Glawion et al., 2023; Mandal, 2024; Valdé and s-Pineda, 2020; Ramalingam et al., 2023; Yang et al., 2021; Pfreundschuh et al.; Antwarg et al., 2021; Ribeiro et al., 2016; Islam et al., 2024).
6. Prediction of Extreme Weather Events and Model Interpretations	(Brotzge et al., 2023; Seo, 2020; Das et al., 2017; Nayak et al., 2013; Tiwari et al., 2018; Lin et al., 2024; Mhedhbi, 2022; Dibike et al., 2006; Song et al., 2024; Lubo et al., 2020; Selvaraju et al., 2020; Satishkumar et al., 2024; Slater et al., 2023)

and robustness, a necessity in NWP due to often limited or noisy observational data (Schultz et al., 2021). Recent studies underscore the efficacy of ML and DL techniques in data augmentation and assimilation within NWP contexts. ML-based data augmentation strategies, such as Generative Adversarial Networks (GANs), generate realistic synthetic data that complement actual observations, thereby increasing training set diversity (Ge et al., 2024). ML algorithms are also helpful in imputing missing values in observational datasets and mitigating data gaps and their impacts (Wangwongchai et al., 2023). DL-augmented data assimilation using convolutional autoencoders has shown improvements in reconstructing missing satellite imagery data (Feng et al., 2021).

Similarly, DL-based augmentation techniques are more advanced than ML. For example, CycleGANs facilitate data translation between different domains, such as converting radar data into satellite imagery, thus expanding the available range of training data (Sun et al., 2023). Autoencoders, TNN, and CNN, are also employed to denoise and reconstruct noisy observational data, significantly enhancing data quality (Feng et al., 2021; Yan et al., 2023; Liguori et al., 2021). These techniques are further extended to the assimilation of unconventional data sources, such as social media and citizen science observations (Rasheed et al., 2019, 2020). Moreover, ML and DL algorithms enable real-time data assimilation, allowing rapid updates to forecast as new data becomes available (Eyre et al., 2022).

Despite the advancements in ML and DL for data assimilation, a significant issue is the computational cost associated with training complex ML and DL models, which often require extensive hardware resources and extended processing times. Additionally, the integration of these algorithms into existing NWP frameworks needs changes in architectures and workflows. Moreover, the black-box nature of many DL models complicates interpretability and transparency, hindering their acceptance by meteorologists who rely on the ability to understand and validate forecast outputs. These advancements suggest that ML and DL have the potential to revolutionize NWP through innovations in data assimilation and augmentation.

4.2. Data pre-processing techniques

After data acquisition, its pre-processing is critical for weather forecasts in NWP (Yang et al., 2016), as shown in Fig. 3. The first pre-processing approach, the Well-mixed Preprocessor (WEMPP), aims to generate an initial estimate of liquid water content, a parameter absent in NAM data (Gough et al.). WEMPP seeks to retain cloud water content, thereby mitigating radiative feedback due to the abrupt removal of the liquid water path (LWP) during initialization. It enhances the micro-physics scheme's ability to establish a stable nocturnal Sc field within 1–2 h (Fay et al., 1997). The second pre-processing scheme, the Cloud Data Assimilation (CLDDA) package developed by Mathiesen et al. (2013), is used at initialization rather than mid-simulation. Both pre-processors offer efficient and computationally inexpensive methods for liquid water initialization tailored for boundary layer clouds, which are critical due to their substantial optical depth (Mathiesen et al., 2013). Recently, researchers employed ML and DL (i.e., RF, CNN, LSTM, etc.), which helps bridge the gap between complex physical models (i.e., NWP, WRF – ARW, or WRF – NMM, etc.) and DDMs for more accurate and interpretable forecasts (Wang et al., 2021; Senocak et al., 2023; Georgakaki et al., 2024; Micolini et al., 2020). WRF model outputs, such as temperature, PPT, WS, PS, and RH, are transformed into informative features that ML/DL models can utilize (Abdulla et al., 2022). These features are combined, filtered, or normalized to capture patterns in atmospheric dynamics. For example, generating lagged PPT captured temporal and spatial dependencies critical for accurate predictions (Maraun et al., 2010). Pre-processing atmospheric data involves interpolation of missing values (Li et al., 2021), detection of outliers (Al- et al., 2024), anomaly detection (Yan et al., 2021), and ensuring high-quality inputs (Shen et al., 2024) for NWP model processing. For example, Recursive Feature Elimination (RFE) (García et al., 2018; Akbarian et al., 2023), XGBoost (Phan et al., 2021), and Principal Component Analysis (PCA) (Gu et al., 2019) are used to refine data for WRF modeling by eliminating non-informative variables. Advanced DL models, including Autoencoders and CNNs (Jia et al., 2021), further optimize feature extraction by capturing complex spatiotemporal relationships.

Data pre-processing in NWP scenarios encounters challenges such as handling incomplete or biased datasets from diverse sources, ensuring consistency across spatial and temporal resolutions, and adapting to rapidly changing atmospheric conditions. Additionally, computational limitations may arise when integrating advanced ML/DL techniques for feature extraction. It requires robust validation to maintain forecast accuracy.

4.3. Quantification of uncertainty

Uncertainty quantification is a process of quantitative classification and inference of uncertainties in real-world and computational applications (Sullivan, 2015). It is a critical component in climate numerical weather modeling (Qian et al., 2016). Uncertainty stems from various sources, such as model failures, input data variability, and atmospheric processes' randomness. It requires robust quantification for reliable forecasts (Mehra, 2021). NWP modeling traditionally employs statistical and probabilistic methods. Statistical techniques such as ensemble

forecasting involve multiple WRF simulations with varied initial conditions or physical parameterizations to estimate forecast uncertainty. Ensemble methods, including Monte Carlo simulations and Bayesian inference, are instrumental in understanding possible outcomes' distribution and associated probabilities (Speich et al., 2021).

Similarly, Bayesian neural networks (BNNs), for example, generate probabilistic forecasts by learning the distribution of model parameters and outputs. It provides a quantifiable measure of confidence (Mitros et al., 2019). In NWP-ML/DL integration, ensembled models, such as GPs and RFs, are also used to estimate uncertainties in weather forecasts. GPs incorporate a probabilistic framework, delivering predictive distributions and confidence intervals derived from training data (Williams et al., 2006). RFs quantify uncertainty through bootstrap aggregation, whereby multiple models are generated from resampled data, and their variability is analyzed to gauge prediction reliability (Breiman, 2001). DL techniques further contribute to uncertainty quantification through methods such as dropout and uncertainty layers. Dropout, initially developed as a regularization tool in neural networks, can be viewed as a Bayesian approximation, generating uncertainty estimates by averaging predictions over multiple stochastic realizations of the network (Gal et al., 2016). Uncertainty-aware architectures, i.e., deep ensembles and probabilistic neural networks, explicitly model prediction uncertainties, enhancing insights into forecast reliability (Tang et al., 2022). NWP fusion with DL enables the assessment of uncertainties inherent in both the simulation and prediction phases, providing a holistic perspective on forecast uncertainty (Cheng et al., 2023).

Despite these advancements in DL for uncertainty quantification, challenges remain. It includes sensitivity to hyperparameters, potential overfitting to training data, and difficulties in interpreting uncertainty estimates. Furthermore, DL models may struggle to accurately capture the complexities of atmospheric processes, which leads to misrepresentation of forecast uncertainties.

4.4. Dynamic tuning of WRF parameters

Dynamic parameter tuning is critical in optimizing NWP models (Ollinaho et al., 2013). To improve the accuracy and efficiency of NWP models, adaptive tuning approaches have been developed due to their complexity and susceptibility to different physical parameterizations and beginning conditions (Loh et al., 1999). In contrast to manual tuning, these strategies offer a more systematic approach by dynamically tuning parameters based on observational data and real-time performance measurements (Ting et al., 2009; Gao, 2003). Traditionally, parameter tuning in WRF has relied on manual adjustments of model parameters, such as grid resolution, boundary conditions, and physical parameterizations. These adjustments are based on empirical experience and trial-and-error methods, which are not automated and do not yield optimal results (Koster et al., 2018). It has led to the development of advanced adaptive tuning methods that incorporate ML/DL methods to optimize model performance. Optimization algorithms such as Genetic Algorithms (GAs) (Sward et al., 2022), Hybrid Particle Swarm Optimization (PSO) (Almaraashi, 2023), and Simulated Annealing (SA) (Zhang et al., 2015) have been widely applied in the adaptive tuning of WRF. These algorithms explore the parameter space and identify optimal configurations by adaptively adjusting parameters based on model performance metrics. GAs, for example, evolve a population of parameter sets over successive iterations, selecting those that provide the best forecast accuracy and convergence (Zhang et al., 2020). PSO and SA also offer better solutions for multi-parameter tuning by iteratively searching for optimal parameter combinations.

ML models replace the traditional NWP modeling optimization tuning method (Couvreur et al., 2021). Traditional models, such as the Kriging method, were replaced with advancements in Gaussian Process models, which approximate the relationship between parameter settings and model performance, enabling efficient parameter space exploration (Christianson et al., 2023). Bayesian optimization extends this approach

with the incorporation of probabilistic models to iteratively update parameter estimates based on observed performance, thereby guiding the tuning process toward optimal configurations (Menon et al., 2020). These techniques allow for more targeted and efficient tuning, reducing the computational burden of running multiple simulations. In the context of WRF-DL models, including reinforcement learning and meta-learning, offer advanced solutions for adaptive tuning (Zhang et al., 2022a; Mohammadi et al., 2020). Reinforcement learning algorithms learn and optimize long-term through parameter adjustment and tuning through interaction with the WRF model (Mnih et al., 2015). Meta-learning, often described as “learning to learn,” leverages knowledge from previous tuning tasks to adjust parameters adaptively, improving the efficiency and effectiveness of the tuning process (Finn et al., 2017).

Real-world applications of dynamic tuning in WRF have shown promising results. For instance, a study by Zhang et al. (2020) utilized Genetic Algorithms to optimize parameter settings for urban weather prediction, significantly improving forecast accuracy in complex urban environments. Similarly, Chen et al. (2021) demonstrated the effectiveness of Bayesian optimization in enhancing rainfall prediction by adaptively adjusting physical parameterizations based on observed data. These case studies highlight the practical benefits of implementing adaptive tuning techniques in diverse meteorological contexts.

4.5. Post-processing techniques

Ensemble forecasting generates multiple simulations to capture uncertainties in weather predictions but often faces biases and spreads calibration issues (Buizza, 2018). Post-processing is intact in refining probabilistic forecasts produced by ensemble models (Grö et al., 2021). Traditional approaches such as Ensemble Model Output Statistics (EMOS) (Baran et al., 2018) and Bayesian Model Averaging (Duan et al., 2021) have been widely adopted to enhance the predictive skill of raw ensemble forecasts by correcting biases and refining uncertainty quantification. Studies by Hamill and Whitaker (2006) have demonstrated the initial application of these methods to re-forecast datasets, targeting temperature at 850 hPa (T850) and geopotential height at 500 hPa (Z500) (Hamill et al., 2007). The recent integration of neural network-based methods into ensemble forecasting represents a significant advancement (Rasp et al., 2018; Brenowitz et al., 2018), emphasizing localized applications at specific weather stations and broader global-scale interpolations. It highlights the expanding potential of ML/DL models in enhancing ensemble forecast accuracy and reliability (Baran et al., 2021).

RF, GB algorithms, and BNNs are used for ensemble forecasts. For example, GB, Extreme GB, RF, and DNN algorithms have been applied to correct daily, seasonal PPT, and water level biases in NWP ensemble outputs, yielding probabilistic forecasts with improved fidelity to observed conditions (Monego et al., 2022; Dong et al., 2023; Lee et al., 2019b). Recent advancements highlight the effectiveness of models such as Quantile Regression Forests and Neural Networks in calibrating probability forecasts for sub-seasonal PPT, snow, and temperatures, thereby enhancing ensemble reliability and reducing forecast errors (Evin et al., 2021; Ahmed et al., 2020; Scheuerer et al., 2020; Ghazvinian et al., 2022). Advanced DL models such as TNNs, CNNs, and GANs have emerged as robust ensemble models. For example, CNNs have been employed for daily PPT forecasts by refining spatial patterns (Wang et al., 2024). GANs have been utilized to generate high-resolution probabilistic forecasts by learning the error distributions inherent in ensemble outputs, resulting in ensemble members that align with observed data (Price et al., 2022; Glawion et al., 2023). Recent applications of TNN models, known for their attention mechanisms, have shown potential in post-processing ensemble forecasts in refining sequential data and capturing long-range temporal dependencies, which are essential for accurately predicting extreme weather events (Mandal, 2024; Vald  and s-Pineda, 2020). BNNs emerged due to their ability to generate probabilistic outputs that explicitly quantify uncertainty, an essential aspect of

effective decision-making in extreme weather forecasting (Mehra, 2021). Incorporating Variational Inference and Monte Carlo Dropout has enhanced the computational efficiency and scalability of BNNs for ensemble post-processing applications (Ramalingam et al., 2023).

Additionally, recalibration techniques, such as Quantile Regression Neural Networks (QRNNs), adjust ensemble spreads to more accurately reflect the inherent uncertainties of aerosol predictions (Yang et al., 2021). By learning the quantiles of observed data, QRNNs provide dynamic adjustments to ensemble forecasts, ensuring that the ensemble spread corresponds to the accurate distribution of forecast uncertainties (Pfreundschuh et al.). The integral complexity of ML/DL models complicates the interpretation of corrected ensemble outputs, underscoring the need for explainable AI (XAI) techniques. XAI approaches, such as Shapley additive explanations (SHAP) (Antwarg et al., 2021) and Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro et al., 2016), are integrated into post-processing workflows to elucidate how ML/DL models modify ensemble forecasts (Islam et al., 2024).

Recent case studies underscore the effectiveness of advanced post-processing techniques in real-time operational forecasting. For example, the application of Quantile Regression Forests at the European Centre for Medium-Range Weather Forecasts (ECMWF) demonstrated substantial improvements in precipitation forecasts, achieving bias reductions of up to 30% over traditional methods (Zhang et al., 2022b). Similarly, the integration of BNNs in the National Weather Service's operations has shown enhanced accuracy in predicting severe weather events, allowing for better risk assessments and decision-making in emergency management scenarios (Mitros et al., 2019). These examples illustrate the transformative impact of ML and DL models in operational settings.

4.6. Prediction of extreme weather events and model interpretations

Extreme weather phenomena are characterized by their high nonlinearity and chaotic nature, such as hurricanes, floods, heatwaves, and severe thunderstorms, which pose challenges for traditional NWP (Brotzge et al., 2023). ML models, i.e., RF, SVMs, and GB, are utilized to predict the occurrence and severity of extreme weather events. These models leverage historical weather data and outputs from the WRF model to identify patterns and make forecasts. For example, by incorporating radar and NWP data into a multimodal ensemble of supervised ML models, RF improves the classification of winter precipitation types, surpassing traditional threshold-based methods (Seo, 2020). Meanwhile, the effectiveness of the RF algorithm in nowcasting convective precipitation events using radiometer brightness temperatures across multiple frequency bands achieved high detection probabilities and low false alarm rates (Das et al., 2017).

Similarly, the SVM algorithm predicts extreme rainfall events in Mumbai by recognizing specific weather patterns and anomalies over a 6–48-h lead time. Incorporation of an anomaly frequency method, the two-phase SVM classifier outperforms traditional statistical techniques in false alarm rates and prediction accuracy (Nayak et al., 2013). A study introduces a short-term solar irradiance forecasting method that combines NWP with GB regression and bootstrap aggregation, incorporating spatial and seasonal parameters. The proposed approach improves forecasting accuracy to address challenges in grid stability and economic operation due to solar energy's intermittency (Tiwari et al., 2018).

DL models, such as CNNs, LSTM networks, TNNs, and GAN models, have revolutionized the prediction of extreme weather events due to their ability to model complex spatial and temporal patterns within atmospheric data. For example, CNNs identified spatial features related to storms and localized events by analyzing high-resolution outputs from the WRF model (Lin et al., 2024). LSTMs are proficient at managing sequential data, making them particularly effective for forecasting time series data associated with extreme weather events such as floods (Mhedhbi, 2022). Similarly, TNNs are used to downscale global climate model outputs to finer local scales for daily precipitation and

temperature in northern Quebec. The TNN method performs better than traditional regression-based statistical models in reproducing observed climate variability and extreme (Dibike et al., 2006). Recently, GANs have created synthetic weather scenarios based on WRF outputs, facilitating realistic simulations of rare but impactful weather events (Song et al., 2024). Such capabilities are instrumental in improving emergency services' preparedness and response strategies.

The interpretability of ML and DL integrated with the NWP model has become crucial. Techniques such as SHAP, LIME, and feature importance analysis have been applied to ML models to provide insights into the factors influencing predictions. For example, SHAP values can elucidate the contributions of specific WRF-derived variables, such as temperature gradients or wind shear, to forecast severe weather (Lubo et al., 2020). Techniques like Grad-CAM (Gradient-weighted Class Activation Mapping) and Integrated Gradients have been developed to visualize the contributions of different input features to the model's predictions, which helps understand CNNs' detection of storm structures and LSTMs' temporal pattern recognition (Selvaraju et al., 2020). Explainable AI (XAI) introduced frameworks tailored to weather and climate models (Satishkumar et al., 2024). These frameworks integrate ML/DL predictions with meteorological domain knowledge to ensure forecasts are both accurate and scientifically sound (Slater et al., 2023). Ongoing research focuses on developing inherently interpretable ML/DL models, such as rule-based systems and attention-based networks, which aim to provide high accuracy and clear insights into their decision-making processes.

5. End-to-end artificial intelligence based forecasting

Previously reviewed studies (Section 4) showed that AI has impacted weather forecasting by improving accuracy and efficiency over different NWP modeling stages. This section reviewed the most recent work trends in AI + NWP hybrid modeling, as given in Table 2, and end-to-end AI-based forecasting.

5.1. AI + NWP hybrid modeling

Many studies integrated various ML and DL models with the WRF model to improve weather predictions across multiple variables. Sayeed et al. (2021) combined CNN with WRF to enhance forecasts for wind speed (WS), wind direction (WD), precipitation (PPT), relative humidity (RH), temperature (T), dew point temperature (DT), and surface pressure (SP) using hourly data from 2014 to 2018. The CNN model reduced biases in WRF predictions, particularly in wind speed, RH, and precipitation, by classification and regression models for rain hours and rainfall amounts (Sayeed et al., 2021). Shirali et al. (2022) explored WRF, ANN, and SVM to forecast precipitation, air temperature (AT), and floods. The study used hourly observations for air pressure (AP), PPT, RH, WS, WD, and river flow and applied the Partial Mutual Information (PMI) algorithm to select influential input variables, with the SVM model proving most accurate in flood prediction (Shirali et al., 2020). Similarly, Zhang et al. (2024) applied a Population-Based Training (PBT) model integrated with a GRU neural network to improve precipitation forecasts. With data collected from 2014 to 2022, the PBT-GRU model significantly outperformed traditional machine learning models like Gradient-Boosted Decision Trees (GBDT), RF, SVM, and KNN (Zhang et al., 2024). Moreover, Cho and Kim (2022) focused on streamflow prediction, integrating the WRF-Hydro model with LSTM networks. Using 3-hourly data from 2013 to 2018, the study found that the WRF-Hydro-LSTM hybrid model outperformed the standard WRF-Hydro model in reducing residual errors in streamflow forecasts (Cho et al., 2022b). Liu et al. (2023) aimed to improve precipitation and runoff forecasts in the Daqing River Basin using a WRF/WRF-Hydro-LSTM approach. With hourly data from 2007 to 2016, their study demonstrated that real-time data assimilation with LSTM improved forecast accuracy (Liu et al., 2023). Wang et al. (2023) developed a hybrid CNN-LSTM model combined with Gamma tests for streamflow prediction, using high-resolution WRF data from 2007 to

Table 2

Review of the hybrid modeling (ML and DL models combined with WRF).

Method	Target Variables	Input Data		Pre-Processing	Post-processing	Objective	Findings	Source
		Obs	WRF					
CNN-WRF	WS, WD, PPT, RH, T, DT, and SP in 2018	Hourly WS, WD, PPT, RH, T, DT, SP in 2014–2017	Hourly WS, WD, PPT, RH, T, DT, SP in 2014–2017	The data was normalized between 0 and 1 to ensure uniformity across all inputs, and the dataset was split into training and validation sets at 50:50.	The study combined a classification model for identifying rain hours and a regression model for predicting hourly rainfall amounts.	Enhance the accuracy (IOA) of weather predictions generated by the WRF model.	The Weather-AI model significantly reduced biases in the WRF model's predictions for WS, RH, and hourly PPT by considering IOA.	Sayeed et al. (2021)
WRF, ANN, and SVM	PPT, air temperature, and flood in March 2016	Hourly PPT, RH, Air Pressure (AP), WS, WD, River flow	Hourly PPT, RH, Air Pressure (AP), WS, WD, River flow (RF)	The study utilized the Partial Mutual Information (PMI) algorithm to identify influential input variables influencing rainfall and temperature predictions. It involves adjusting model parameters based on historical data to improve prediction accuracy.	Model outputs evaluation, simulating hydrographs, and comparing model performances.	Forecast PPT, air temperature, and flood with the best model from WRF, ANN, and SVM	The comparison between the peak discharge and run-a-height value of the predicted flood in different models showed that the SVM model had more efficiency and accuracy than the other models in predicting PPT, AT, and flood.	Shirali et al. (2020)
PBT-GRU, RF, SVM, KNN, and Gradient-Boosted DT (GBDT), and WRF	PPT	Hourly AP, visibility, WD, WS, AT, RH, PPT, Radar Reflectivity factor (NWP)	Hourly AP, visibility, WD, WS, AT, RH, and PPT	Data preparation, Data processing, Sample database construction, and Dataset division	Model evaluation, comparison with other models, and Model adjustments	This study aims to enhance the accuracy of hourly precipitation forecasts, which apply ML to improve WRF	The results indicated that the PBT-GRU model significantly outperformed traditional ML algorithms (RF, SVM, KNN, GBDT) in correcting PPT forecasts.	Zhang et al. (2024)
WRF-Hydro, LSTM, and WRF-Hydro-LSTM	Streamflow	3-h Specific humidity (SH), Incoming longwave radiation (ILR), Near-surface wind (NW), AT, SP, PPT Precipitation rate (PPT _r) and Observed inflow	3-h SH, ILR, NW, AT, SP, PPT PPT _r , Residual error, and Observed inflow	Data Collection and Residual Error Prediction by LSTM	Model evaluation and Sensitivity analysis	Enhance the accuracy of streamflow predictions by integrating the WRF-Hydro with LSTM networks.	The results indicated that the WRF-Hydro-LSTM model significantly outperformed the WRF-Hydro-only model.	Cho et al. (2022b)
WRF/WRF-Hydro and LSTM	PPT and runoff	Hourly AT, pressure (P), WS, humidity, and PPT	Hourly AT, pressure (P), WS, humidity, and PPT	Data Conversion and Assimilation Method	Forecast Evaluation, Runoff Forecasting, and Evaluation	The primary objective of the study was to improve the accuracy of precipitation and runoff forecasts in the Daqing River basin	The study demonstrated that the WRF/WRF-Hydro coupled system, enhanced by real-time data assimilation by LSTM, improved precipitation and runoff forecasts.	Liu et al. (2023)
Gamma Test, WRF, and CNN-LSTM	Streamflow	Daily Surface skin temperature, SP, SH, PPT, Soil moisture (0–10, 10–40, 40–100 cm), Windspeed, and Streamflow All datasets from WRF	Global analysis from Final Operational Global Analysis (FNL)	Data segmentation, Variable selection, and Input matrix formation	Model evaluation, Performance improvement, and Application of Results	By leveraging high-resolution meteorological data to improve streamflow predictions, develop a hybrid CNN-LSTM for simulating daily streamflow in two mountainous	1. The results indicated that the CNN-LSTM model performed well in simulating streamflow using high-resolution WRF data. 2. The application of the Gamma test improved the	Wang et al. (2023)

(continued on next page)

Table 2 (continued)

Method	Target Variables	Input Data		Pre-Processing	Post-processing	Objective	Findings	Source
		Obs	WRF					
CNN, RNN, LSTM and WRF	WS	Daily WS from WRF	Hourly WS	Handling missing values, Interpolation, Data resampling, and Normalization	Model evaluation and Performance improvement	catchments of the Daqing River Basin, northern China. Develop hybrid CNN-LSTM for simulating daily WS	model's performance Improving upon the WRF model, they show that the combination of CNN and RNN produces day-ahead wind speed predictions of high accuracy, with no need for previous measurements.	Christoforou et al. (2023)
DeepWIA and WRF-Chem	PM2.5 concentrations	Daily averaged meteorological data (ERA5)	Meteorological Data were (to define initial boundary) PM 2.5, PM 10, SO ₂ , NO ₂ , O ₃ , CO, and Multi-resolution Emission Inventory for China (MEIC)	Model evaluation and Performance improvement	Capture synoptic-scale variations in aerosol pollution with higher accuracy than WRF-Chem.	Outperformed WRF-Chem and other ML/DL methods in capturing aerosol variations across cities in China with better R2 and RMSE values.		Feng et al. (2023)
FLEXPART-WRF	Surface deposition of airborne contaminants	WRF multi-physics ensembles	Meteorological Data and Physics Parameter (such as cumulus convection, boundary layer turbulence, and land surface interactions)	Statistical emulator for spatial deposition	Estimate effects of weather uncertainty in the dispersion of airborne contaminants.	The statistical model efficiently captured uncertainty without running full ensembles, offering a significant speedup over traditional methods.		Gunawardena et al. (2021)
WRF-Chem and ANN Models	PM2.5, AOD	Meteorological data and aerosol emissions	West Asia Source Function (WASF), Meteorological Data, Satellite Observations, and Aerosol Optical Depth (AOD)	Model evaluation, Error Analysis, and Incorporation of additional predictors	Forecast aerosol concentrations and AOD using deterministic and statistical models.	WRF-Chem simulations were enhanced with ANN models, improving the accuracy of PM2.5 and AOD forecasts.		Nabavi et al. (2018)

2019. Their model simulated streamflow, and the Gamma test further refined the predictions (Wang et al., 2023). Christoforou et al. (2023) also focused on WS forecasting, utilizing a hybrid CNN-RNN model based on hourly wind speed data from 2016 to 2019. Their study showed that the CNN-RNN model provided highly accurate day-ahead WS predictions without needing prior wind observations, outperforming the conventional WRF model. Similarly, DeepWIA uses daily averaged meteorological data to capture synoptic-scale fluctuations in particulate matter 2.5 µm (PM_{2.5}) concentrations. It outperforms the traditional WRF-Chem and other ML/DL models across Chinese cities (Feng et al., 2023). FLEXPART-WRF, which employs WRF Multiphysics ensembles, adds meteorological uncertainty into airborne pollutant dispersion estimations, considerably increasing computing efficiency through statistical simulation (Gunawardena et al., 2021). Another study enhances PM_{2.5} and AOD forecasts using WRF-Chem and Artificial Neural Networks (ANN), highlighting the potential of combining ML approaches with deterministic models to improve predictive accuracy (Nabavi et al., 2018).

5.2. End-to-end AI-based weather and climate forecasting

In recent years, AI models have shown promise in real-time operational forecasting, demonstrating significant advancements in predictive accuracy. However, their performance is influenced by computational efficiency and scalability challenges. The computational demands of complex models can hinder their real-time application, particularly when

processing vast datasets. Additionally, scaling these models for extensive geographic areas requires careful consideration of resource allocation and algorithmic optimization. Recent studies have illustrated successful implementations of AI models in operational settings, highlighting best practices for enhancing computational efficiency while maintaining prediction quality.

Despite the advancements in AI modeling, as discussed in the previous section, no study is still available on end-to-end AI-based forecasts. However, some of the most recent studies tried to refine AI models to forecast weather and climate but depended on WRF models, such as the predictive RNN (PredRNN), which incorporates additional connections between adjacent time steps within a core stacked spatiotemporal LSTM (ST-LSTM) framework. PredRNN utilizes a dual-memory mechanism to simultaneously encode temporal and spatial variations within a unified memory pool. Variants such as PredRNN++ have demonstrated the model's versatility and efficacy in precipitation nowcasting (Wang et al., 2022b). Google Research has also contributed notable advancements with high-resolution precipitation nowcasting models. These include U-Net CNNs to frame forecasting as an image-to-image translation problem and the MetNet model, which employs axial self-attention mechanisms to outperform NWP models in specific temporal and spatial scales (Han et al., 2021). MetNet represents a significant milestone in DL-based weather forecasting, setting new benchmarks in performance.

Traditional AI models often train on low-resolution data, overlooking small-scale dynamics and terrain impacts. To address these challenges,

NVIDIA developed FourCastNet, which utilizes the Adaptive Fourier Neural Operator (AFNO) to achieve high-resolution (0.25°) forecasts, significantly outperforming NWP models in speed and resolution (Kurth et al., 2023). DeepMind's GraphCast leverages graph neural networks (GNNs) to accurately predict the weather 10 days ahead, surpassing the ECMWF operational forecast system (Lam et al., 2023). FengWu and FuXi models tackle long-term forecasts using advanced multimodal (Shu et al., 2024). NowcastNet integrates DL algorithms with physical laws to model mesoscale and convective precipitation processes, enhancing predictive performance across key indices (Zhang et al., 2023). Similarly, ClimateNet, a DL model based on expert-labeled data, has been employed to detect high-precision analyses of extreme weather (Prabhat et al., 2020). Following TECA analysis, a deep CNN (NeXTNow) was developed to predict weather radar data for nowcasting purposes (Albu et al., 2022). A hybrid CNN-LSTM model has been developed for typhoon intensity and formation forecasting, combining a 3D CNN for spatial feature extraction with an LSTM for temporal correlation analysis (Chen et al., 2019). A recent Pattern Analysis and Applications study investigated the integration of TCN with LSTM layers. The model demonstrated effectiveness in multi-input single-output and multi-input multi-output regression tasks, showing promise for accurate and efficient forecasting up to 12 h ahead, surpassing established models such as the WRF-NWP model (Hewage et al., 2021). Pangu-Weather, trained on 39 years of global data, has accurately surpassed the ECMWF operational integrated forecasting system. The system consists of 3D DNNs and hierarchical temporal aggregation strategies that effectively address complex weather patterns and accumulation errors, enhancing its performance in ensemble and extreme weather forecasts (Bi et al., 2023).

These studies show that AI can improve forecasting accuracy. These cutting-edge models illustrate the integration of AI-based models with NWP, highlighting a collaborative approach where AI expands NWP outputs but relies on NWP for initial state determination. This comprehensive study emphasizes the complementary roles of AI and NWP in advancing weather forecasting, with reanalysis data playing a crucial role in training sophisticated ML/DL models.

5.3. Limitations of end-to-end AI-based models

AI has transformed weather forecasting, but it has some limitations. One significant limitation is the weak interpretability of AI models, which are often criticized as “black boxes” due to their lack of transparency in decision-making processes (Waqas et al., 2023; Brotzge et al., 2023). From the reviewed studies, the following are the limitations of AI-based modeling.

- Interpretability is crucial in weather forecasting and climate prediction as it enhances model credibility, improves decision support, and provides insights into meteorological phenomena. Although some research has addressed the explainability of AI models, no studies currently focus on data-driven global weather forecasting models. The lack of interpretability hampers understanding of the model's reasoning, making it challenging to refine methods or trust predictions in critical applications where decisions impact lives and infrastructure.
- Another key limitation is the high reliance on high-quality training data. AI-based weather models often use datasets generated by NWP, which can limit the models' learning effectiveness and generalization ability. The extent and quality of the data, such as the number of variables, vertical levels, and resolution, constrain these models' predictive capabilities. Data assimilation techniques, which integrate diverse observational data into a comprehensive atmospheric framework, still face challenges in handling high-resolution and high-dimensional data. The precision of these methods is essential, but current approaches are often insufficient to manage the complexity of modern datasets effectively.

- Uncertainty quantification remains a critical aspect of AI-based forecasting. To address this, statistical post-processing methods can be employed to mitigate error accumulation over time. Techniques such as Bayesian inference and ensemble forecasting provide valuable frameworks for estimating predictive uncertainty. By integrating these approaches, researchers can improve the robustness of AI models, enabling more reliable long-term forecasts. This expanded belief in uncertainty quantification strategies is essential for enhancing the credibility of AI predictions in weather and climate applications.
- Moreover, end-to-end AI-based models often perform unsatisfactorily in extreme weather cases due to the lack of a comprehensive description of such conditions in their training data. AI models typically generate smoothed outputs, which can underestimate the magnitude of extreme events. Even though specific models like Pangu, GraphCast, and NeuralGCM have shown promising results in extreme cases, they still fall short in systematically evaluating and managing extreme weather scenarios, leading to potential underestimation of risks associated with severe weather.
- The evaluation of data-driven weather models is often incomplete. Current assessments primarily use metrics like RMSE, ACC, and CRPS, which may not fully capture a model's effectiveness, especially when conducted by data scientists who lack meteorological evaluation expertise. Establishing a standardized, comprehensive evaluation benchmark is crucial to ensure the reliability of these models.

These limitations of end-to-end AI-based models underscore the need for further research and development to improve their interpretability, data dependence, uncertainty management, and robustness in extreme weather forecasting.

6. Conclusion and recommendations

This survey offers a technical review of the integration of AI with NWP models across various weather and climate modeling stages. It underscores the potential of AI-based models to enhance forecasting accuracy. Despite notable advancements, the surveyed literature indicates that current AI models function as supplementary tools, extending NWP outputs through post-processing, data assimilation, and pre-processing techniques. The evolution of AI models—including RNNs, CNNs, LSTMs, and other architectures such as MetNet, FourCastNet, and GraphCast—demonstrates their capability to manage high-resolution forecasts and address specific weather phenomena. However, significant gaps persist in the development of end-to-end AI-based weather forecasting models. Challenges such as limited interpretability, high dependency on NWP-generated training data, difficulties in uncertainty quantification, and suboptimal performance in extreme weather events continue to constrain the full potential of AI in weather prediction.

This survey proposes specific actions: integrate high-resolution datasets, enhance model interpretability, and adopt advanced training techniques focused on extreme conditions. These insights can improve forecasting accuracy and enable better preparedness and response strategies for mitigating the impacts of severe weather events. Future research should prioritize the development of interpretable AI models to enhance credibility and transparency. Improving data quality and resolution through advanced data assimilation techniques is essential for maximizing the generalization capabilities of AI models. Additionally, robust uncertainty quantification methods must be developed to address variability and error propagation in AI-driven forecasts. Special emphasis should be placed on enhancing AI models' ability to accurately predict extreme weather events by incorporating comprehensive descriptions of such conditions during training.

CRedit authorship contribution statement

Muhammad Waqas: Writing – original draft, Visualization,

Investigation, Conceptualization. **Usa Wannasingha Humphries:** Supervision, Resources, Project administration, Funding acquisition. **Bunthid Chueasa:** Writing – review & editing, Investigation. **Angkool Wangwongchai:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of Abbreviations

IoT	Internet of Things
DA	Data assimilation
EnVar	The ensemble–variational
EFI	Extreme forecast index
JEDI	Assimilation integration
ML	Machine learning
DL	Deep learning
GPUs	Graphical processing units
CNNs	Convolutional neural networks
GPUs	Graphical processing units
DNN	Deep neural networks
PBT-GRU	Population-based training gated recurrent unit
RMSE	Root mean square error
RQs	Research questions
NWP	Numerical weather prediction
MDMs	Model-driven models
ARMA	Autoregressive moving average
ARIMA	Autoregressive integrated moving average
SARIMA	Seasonal autoregressive integrated moving average
VAR	Vector autoregression
LSTM	Long short-term memory
GRU	Gated recurrent units
RNNs	Recurrent neural networks
VAE	Variational auto-encoders
GAN	Generic adversarial networks
GFS	Global Forecast System model.
NAM	North American model.
ECMWF	European medium-range forecast model.
CMC	Canadian Meteorological Center model.
RAP	Rapid refresh.
HRRR	High-resolution rapid refresh.
WRF-ARW	Advanced Research Weather Research and Forecasting.
WRF-NMM	Weather Research and Forecasting Nonhydrostatic Mesoscale Model.
NAM NEST	North American mesoscale model conus nest.
WS	Wind speed
WD	Wind direction
PPT	Precipitation
RH	Relative humidity
T	Temperature
DT	Dew point temperature
SP	Surface pressure
AT	Air temperature
PMI	Partial mutual information
PBT	Population-based training
GRU	Gated recurrent unit
GBDT	Gradient-boosted decision trees
P	Pressure
PPT _r	Precipitation rate
ILR	Incoming longwave radiation
NW	Near-surface wind
SH	Specific humidity
FNL	Final operational global analysis
SVD	Singular Value Decomposition
PCA	Principle Component Analysis
ICA	Independent Component Analysis
SVM	Support Vector Machine
WNN	Wavelet Neural Network
CNN	Convolutional Neural Network
DNN	Deep Neural Network
RNN	Recurrent Neural Network
TCN	Temporal Convolutional Neural

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GANs	Generative Adversarial Networks
WEMPP	Well-mixed Preprocessor
CLDDA	Cloud Data Assimilation
PCA	Principal Component Analysis
EMOS	Ensemble Model Output Statistics
BNNs	Bayesian Neural Networks
QRNNs	Quantile Regression Neural Networks
XAI	explainable AI
SHAP	Shapley additive explanations
LIME	Local Interpretable Model-agnostic Explanations
GPs	Gaussian Processes
Gas	Genetic Algorithms
PSO	Particle Swarm Optimization
SA	Simulated Annealing
Grad-CAM	Gradient-weighted Class Activation Mapping

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